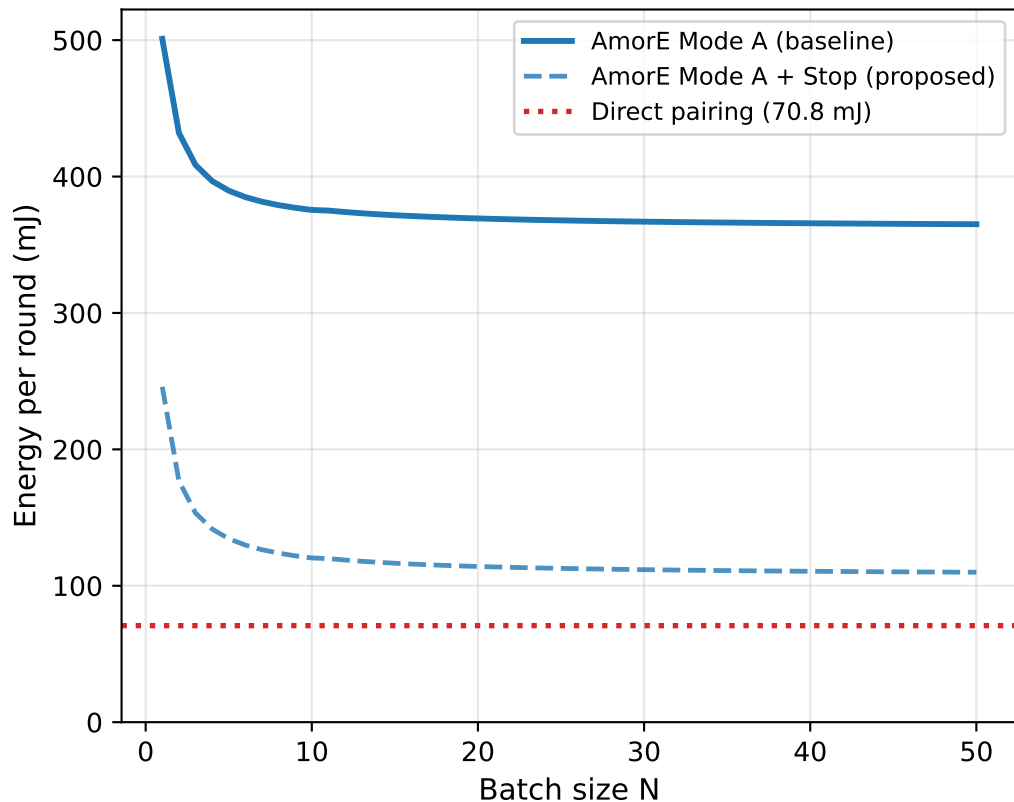
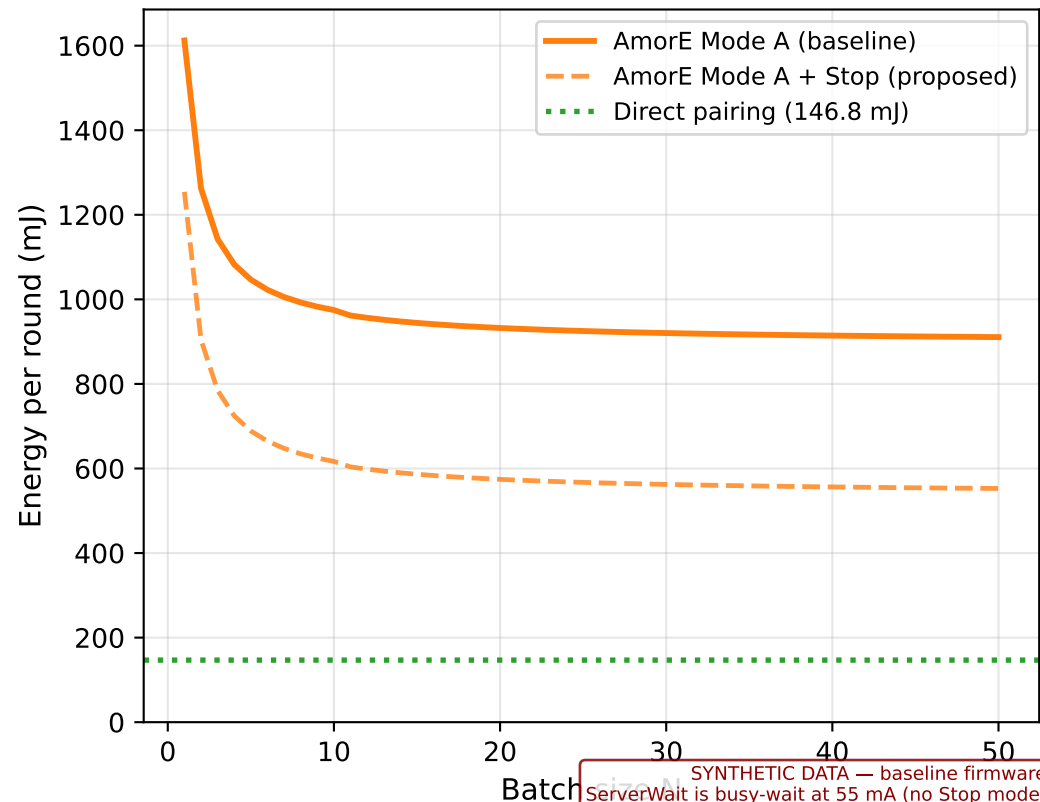


Figure 1: Energy per round vs batch size N on STM32F407 Cortex-M4
 BN254: AmorE costs 1.51× a single direct pairing. BLS12-381: AmorE costs 1.22× three direct pairings (the closer analogue, since each AmorE round contains three pairing-equivalent verification steps on the server). Trade-off: 33× less SRAM, 3× less Flash, input privacy, cheating detection.

BN254



BLS12-381



SYNTHETIC DATA — baseline firmware
 ServerWait is busy-wait at 55 mA (no Stop mode)
 See ADR-005, ADR-009